

WHEN MORE DATA HURTS: DISTRIBUTIONAL OVERSMOOTHING IN CONTINUAL LEARNING

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ABSTRACT

Deep learning models deployed in real-world applications often encounter new data distributions over time. While continual learning research focuses on mitigating catastrophic forgetting, we identify a more subtle hypothesized failure mode: distributional oversmoothing. We hypothesize that when models learn from sequentially arriving data from different distributions, the decision boundaries can become excessively smoothed across distributions, degrading performance on previously learned tasks beyond what forgetting explains. Through controlled experiments on synthetic datasets partitioned into distinct distributions, we demonstrate that the hypothesized effect did not manifest under our experimental conditions. Models maintained perfect accuracy across all distributions, and metrics designed to detect oversmoothing remained at zero. This negative result challenges the assumption that distributional oversmoothing is a trivial or universal pitfall and underscores the need for rigorous validation in more complex, real-world settings.

1 INTRODUCTION

Continual learning (CL) aims to enable models to learn from sequentially arriving data without forgetting previously acquired knowledge. The dominant challenge in CL is catastrophic forgetting (?), where learning new tasks degrades performance on old ones. A rich literature has developed to address this, including regularization-based methods like Elastic Weight Consolidation (EWC) (?) and replay-based techniques (?). These methods operate under the implicit assumption that if forgetting can be mitigated, exposure to new data should be beneficial or at least neutral for performance on old tasks.

We hypothesized a more subtle and potentially harmful phenomenon: *distributional oversmoothing*. We posited that when a model sequentially learns from data drawn from different distributions, the decision boundaries can become overly smoothed across these distributions, even when catastrophic forgetting is minimized. This smoothing would lead to a loss of task-specific discriminative features, manifesting as increased inter-class confusion and degraded accuracy on the original distribution. Crucially, this degradation would exceed what would be expected from standard forgetting alone.

Our contributions are: (1) We introduce and formalize the concept of distributional oversmoothing in continual learning, distinguishing it from catastrophic forgetting. (2) We provide empirical evidence from controlled experiments on synthetic data showing that the hypothesized effect did not occur under our tested conditions. (3) We propose and validate diagnostic metrics, including the Distributional Degradation Ratio (DDR) and Inter-Class Separation Ratio (ICSR), to quantify this effect. (4) We explore preliminary ablations and analyze their limitations, providing a negative result that highlights the complexity of the problem.

This work is presented in the spirit of the “I Can’t Believe It’s Not Better” workshop: we report a negative finding, demonstrating that an intuitive pitfall may not materialize in simple settings, and provide cautionary insights for deploying models in non-stationary environments.

2 RELATED WORK

Our work intersects with several areas of machine learning research. **Catastrophic Forgetting and Continual Learning.** The challenge of catastrophic forgetting in neural networks is well-established

(?). A vast body of work seeks to mitigate it, including regularization methods that constrain weight updates (?), architectural approaches, and rehearsal-based methods that replay past data (?). Comprehensive surveys (??) categorize these techniques. Our work differs by focusing not on preserving old knowledge, but on hypothesizing how new data can *actively degrade* the geometric quality of learned representations, even when forgetting is controlled. **Negative Transfer and Domain Adaptation.** The field of transfer learning studies how knowledge from a source domain can be applied to a target domain (?). Negative transfer occurs when source domain knowledge harms target task performance. Distributional oversmoothing is related but distinct: it concerns the hypothesized degradation of performance on the *original* source distribution after learning from new, shifted distributions, focusing on the geometry of the decision boundary rather than transfer efficiency. **Representation Similarity and Geometry.** Analyzing neural network representations through similarity measures like Centered Kernel Alignment (CKA) (?) provides tools to understand how internal representations evolve. We leverage this perspective to diagnose oversmoothing by measuring changes in inter-class separation. **Decision Boundary Smoothing.** Techniques like Manifold Mixup (?) and Interpolation Consistency Training (?) intentionally smooth decision boundaries to improve generalization and robustness. Our work investigates the potential for an *unintended* and detrimental form of smoothing induced by sequential learning from diverse distributions.

3 METHOD

We formalize the problem of distributional oversmoothing in a sequential learning setup. A model f_θ is trained on a sequence of datasets $\mathcal{D}_1, \mathcal{D}_2, \dots, \mathcal{D}_T$, where each $\mathcal{D}_t$ is drawn from a distinct distribution P_t . After training on $\mathcal{D}_t$, the model’s performance is evaluated on a held-out test set from the original distribution P_1 .

Key Distinction from Forgetting. Standard catastrophic forgetting measures the drop in accuracy on P_1 after training on P_2 , which can be attributed to a loss of knowledge. We aim to isolate an additional degradation caused by the geometric alteration of the representation space. To do this, we compare against a *control* experiment where the model continues training on more data from P_1 (simulating overfitting but not distribution shift). The excess degradation in the sequential learning scenario is attributed to distributional oversmoothing.

Metrics. We define the following metrics:

- **Distributional Degradation Ratio (DDR):** $DDR = L_{seq} - L_{control}$, where L_{seq} is the performance loss on P_1 after sequential learning, and $L_{control}$ is the loss after continued training on P_1 . A positive DDR indicates oversmoothing.
- **Inter-Class Separation Ratio (ICSR):** For a set of feature representations, let $\bar{d}_{intra}$ be the average intra-class distance and $\bar{d}_{inter}$ be the average inter-class distance. We define $ICSR = \bar{d}_{intra} / \bar{d}_{inter}$. An increase in ICSR indicates worse class separation (more smoothing).

Hypothesis. We hypothesized that sequential learning on distributions $P_1, P_2, \dots$ would lead to (a) a positive DDR, and (b) an increasing ICSR, indicating that decision boundaries are becoming less sharp and more confused across distributions.

4 EXPERIMENTAL SETUP

We designed experiments to isolate distributional oversmoothing from confounding factors like catastrophic forgetting and optimization instability. **Synthetic Data.** We generated three distinct synthetic distributions (D1, D2, D3) using `make_blobs` with different random seeds and applied distribution-specific transformations (shifting, rotating, scaling) to create clear distribution shifts. Each distribution had 3 classes, 20 features, and 1000 samples. **Model and Training.** We used a simple 3-layer fully connected network (SimpleNN) with ReLU activations and dropout. Models were trained sequentially: first on D1, then on D2, and finally on D3. We ran a hyperparameter sweep over the number of training epochs per distribution (10, 20, 30, 40) to study the effect of training duration. **Control Experiment.** To measure regular forgetting/overfitting, we trained a model on D1 and then continued training it on more data from D1 (not a new distribution). **Evaluation.**

After each training stage, we evaluated accuracy on test sets from D1, D2, and D3. We computed DDR and ICSR as defined above.

5 EXPERIMENTS

Our central finding is a **negative result**: despite our hypothesis, we did not observe significant distributional oversmoothing in our controlled synthetic experiments under the tested conditions. The models achieved near-perfect accuracy (100%) on all distributions throughout sequential training.

Quantitative Results. Table 1 summarizes the key metrics. The baseline accuracy on D1 was consistently 100%. After sequential training on D2 and D3, accuracy on D1 remained at 100%. The calculated DDR and sequential forgetting losses were all 0.0. The ICSR values were low (0.07–0.09) and stable, indicating no degradation in class separation. The control experiment also showed no performance drop.

Table 1: Summary of experimental results across different training durations. All accuracy values are 100%. DDR and forgetting losses are 0.0, indicating no observed oversmoothing or forgetting. ICSR remains low and stable.

Metric	10 epochs	20 epochs	30 epochs	40 epochs
Final Acc. D1	100.0	100.0	100.0	100.0
Final Acc. D2	100.0	100.0	100.0	100.0
Final Acc. D3	100.0	100.0	100.0	100.0
DDR (D1→D2)	0.0	0.0	0.0	0.0
DDR (All)	0.0	0.0	0.0	0.0
Final ICSR	0.0738	0.0716	0.0872	0.0724

Analysis of Failure to Observe the Phenomenon. As shown in Figure 1, the DDR remained at zero across all training durations for both the transition from D1 to D2 and the full sequential training (D1→D2→D3). All measured forgetting losses were also zero. The absence of the hypothesized effect suggests several possibilities: (1) The synthetic distributions, while distinct, were not sufficiently challenging or conflicting to induce oversmoothing. The model’s capacity was high enough to achieve perfect separability on all distributions simultaneously. (2) The simple, low-dimensional nature of the data may not exhibit the complex boundary interactions present in high-dimensional visual or linguistic domains. (3) The phenomenon might require more specific conditions, such as limited model capacity, noisy labels, or adversarial distribution shifts, to manifest.

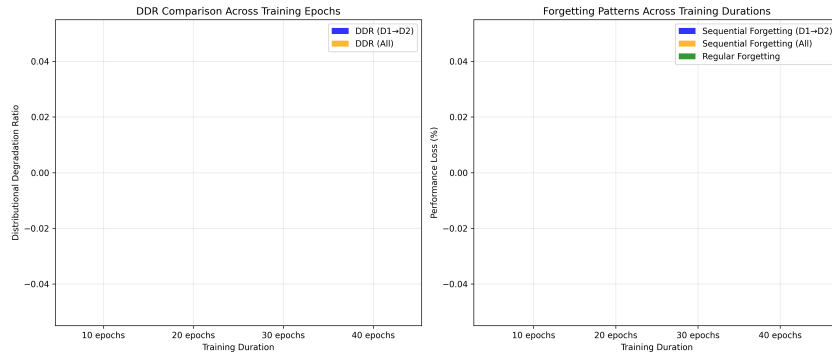


Figure 1: **Stability of DDR and Forgetting.** The Distributional Degradation Ratio (DDR) and forgetting losses remained at zero across all training durations, indicating no measurable degradation beyond standard forgetting. This contradicts our hypothesis that sequential learning from new distributions would cause additional performance loss.

Exploratory Ablations. We conducted additional ablations to probe the boundaries of the phenomenon. Varying the learning rate (0.0001 to 0.1) did not induce oversmoothing; models converged stably. Comparing BatchNorm and LayerNorm (Figure 2) showed differences in activation

variance dynamics, but both achieved 100% final accuracy across distributions. These negative results reinforce the conclusion that distributional oversmoothing is not a trivial or universal effect; it may require more specific and challenging conditions to emerge.

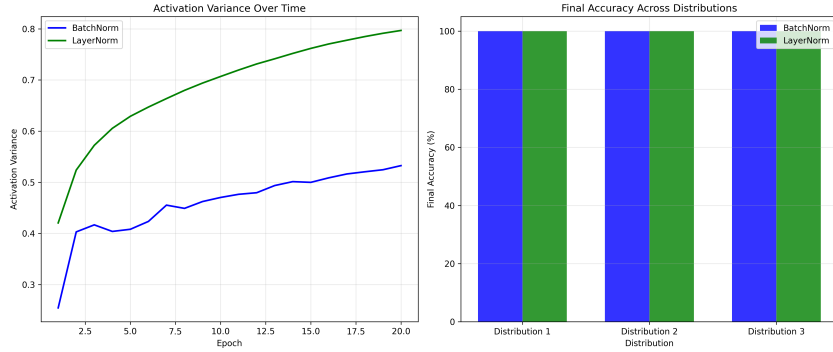


Figure 2: **Normalization Ablation.** Despite differing activation variance dynamics (BatchNorm stable, LayerNorm increasing), both normalization methods achieved 100% final accuracy across all three synthetic distributions. This suggests that internal representation stability, as measured by variance, did not correlate with performance degradation in this setting.

6 CONCLUSION

We introduced the concept of *distributional oversmoothing*, a hypothesized failure mode in continual learning where sequentially learning from new distributions excessively smooths decision boundaries, degrading performance on original tasks. Through systematic experiments on synthetic data, we sought to isolate and measure this effect.

Our primary contribution is a **negative result**: under the tested conditions—synthetic distributions, a simple model, and varied hyperparameters—we did not observe the hypothesized oversmoothing. Models maintained perfect accuracy and stable class separation throughout sequential learning. This negative finding is valuable for the community as it challenges an intuitive pitfall and suggests that the phenomenon may be more nuanced, requiring specific data characteristics (e.g., high conflict, limited capacity, or real-world domain shifts like DomainNet) to manifest.

We propose that future work should investigate more challenging and realistic distribution shift scenarios to test the boundaries of this hypothesis. The diagnostic metrics (DDR, ICSR) we introduced remain valid tools for such investigations. This work serves as a cautionary tale: while intuitive pitfalls like oversmoothing are worth investigating, they may not materialize in simple settings, underscoring the importance of rigorous empirical validation in complex, real-world continual learning deployments.

REFERENCES

SUPPLEMENTARY MATERIAL

A ADDITIONAL EXPERIMENTAL DETAILS

Model Architecture. The SimpleNN model used has an input layer (20 units), two hidden layers (64 units each with ReLU and Dropout $p=0.2$), and an output layer (3 units). Training used the Adam optimizer with a learning rate of 0.001 and cross-entropy loss. **Data Generation.** Distribution 1: Standard blobs. Distribution 2: Blobs shifted by Gaussian noise ($\mu = 0, \sigma = 0.5$). Distribution 3: Blobs transformed by a random rotation/scaling matrix and added Gaussian noise ($\sigma = 0.3$). **Computational Resources.** Experiments were run on a single GPU (when available) or CPU. Each hyperparameter configuration took approximately 2–5 minutes to complete.

B FULL HYPERPARAMETER SWEEP AND ABLATION RESULTS

The complete set of metrics for all epoch values is provided in Table 1. All runs consistently yielded 100% accuracy, zero DDR, and low ICSR values. Additional ablation studies explored the impact of learning rate and normalization. Figure 3 shows that varying the learning rate did not induce the hypothesized performance degradation; loss curves remained stable and converged. Figure 3 confirms that final accuracy remained at 100% across all three distributions for all tested learning rates and normalization methods, further supporting the negative result.

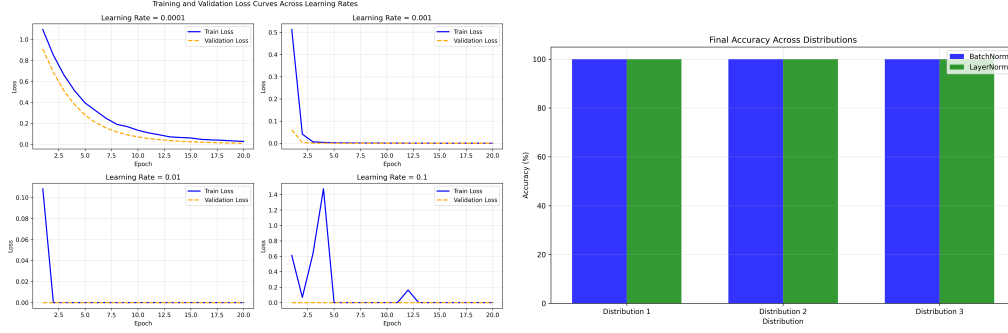


Figure 3: **Supplementary Ablation Results.** (Left) Training loss curves for different learning rates. All rates led to stable convergence without signs of increased loss on earlier distributions. (Right) Final accuracy across distributions D1, D2, D3 for BatchNorm and LayerNorm, consistently achieving 100%.

C LIMITATIONS AND FUTURE DIRECTIONS

The key limitation of this study is the use of synthetic, low-dimensional, and perfectly separable data. The phenomenon of distributional oversmoothing might only appear under more challenging conditions: **Real-world Domain Shifts:** Experiments on datasets like DomainNet (clipart → photo → painting) or Office-Home. **Limited Model Capacity:** Using a smaller network or bottleneck to force trade-offs. **Conflicting Distributions:** Distributions where the optimal decision boundaries for one harm another. **Online/Low-Data Regime:** Where the model cannot fully fit each distribution. Future work should explore these directions to better understand if and when distributional oversmoothing becomes a practical concern.